

Evaluating Decoder-Only and Encoder-Based Language Models on Vietnamese Hate Speech Classification

1. Introduction

1.1 Problem Statement

- Tổng quan về hate speech detection trong tiếng Việt
- Thách thức:
 - ngôn ngữ không chuẩn,
 - slang,
 - implicit toxicity,
 - dữ liệu mất cân bằng

1.2 Research Motivation

- Sự phát triển của các kiến trúc Transformer:
 - encoder-only,
 - encoder-decoder,
 - decoder-only
- Thiếu benchmark toàn diện trên tiếng Việt

1.3 Research Objectives

- Tái lập benchmark từ ViHateT5: Enhancing Hate Speech Detection in Vietnamese With a Unified Text-to-Text Transformer Model
- So sánh hiệu năng giữa nhiều kiến trúc language model
- Đánh giá khả năng semantic understanding của decoder-only trên tiếng Việt

1.4 Research Questions

- Encoder-only có còn vượt trội trong classification task?
 - Decoder-only có hiểu semantic toxicity tốt hơn không?
 - Prompt-based classification có hiệu quả trên tiếng Việt không?
-

2. Related Work

2.1

2.2 Vietnamese Pretrained Language Models

- PhoBERT
- ViGPT
- ViSoBERT / ViSoLex
- ViHateT5

2.3 Transformer Architectures

Encoder-Only

- Bidirectional contextual understanding
- Classification-oriented representation learning

Encoder-Decoder

- Unified text-to-text framework

Decoder-Only

- Autoregressive language modeling
 - Prompt-based classification
 - Label probability inference
-

3. Dataset Construction and Preprocessing

3.1 Dataset Reproduction

- Tái lập 3 dataset từ ViHateT5 benchmark
- Chuẩn hóa dữ liệu và label schema

3.2 Data Analysis

- Class distribution
- Dataset statistics
- Linguistic characteristics

3.3 Data Preprocessing

- Text normalization
 - Unicode normalization
 - Emoji, URL, slang handling
 - Tokenization pipeline
-

4. Methodology

4.1 Encoder-Only Models

- PhoBERT
- ViSoBERT
- Fine-tuning with classification head

4.2 Encoder-Decoder Models

- ViHateT5
- Text-to-text classification framework

4.3 Decoder-Only Models

- PhoGPT
- Mistral 4B
- Prompt engineering
- Constrained label prediction
- Probability-based classification

4.4 Experimental Standardization

- Unified evaluation setting
 - Prompt template design
 - Hyperparameter configuration
 - Training strategy
-

5. Experimental Setup

5.1 Environment Configuration

- Hardware specification
- Software framework
- Training infrastructure

5.2 Evaluation Metrics

- Accuracy
- Precision
- Recall
- Macro F1-score
- Per-class F1-score

5.3 Benchmark Protocol

- Cross-model comparison
 - Cross-architecture comparison
 - Statistical evaluation
-

6. Results and Discussion

6.1 Quantitative Results

- Overall benchmark results
- Architecture-level comparison

6.2 Error Analysis

- Implicit hate speech
- Sarcasm and contextual toxicity
- Slang and informal language

6.3 Decoder-Only Analysis

- Label probability behavior
- Prompt sensitivity
- Output stability

6.4 Computational Analysis

- Training cost
 - Inference latency
 - Memory consumption
-

7. Conclusion and Future Work

7.1 Conclusion

- Tổng kết findings chính
- Đánh giá hiệu quả từng kiến trúc

7.2 Limitations

- Dataset limitations
- Resource constraints
- Generalization issues

7.3 Future Directions

- Larger multilingual SLMs
- Instruction tuning
- Context-aware hate speech detection
- Cross-domain evaluation

VIHATET5: Enhancing Hate Speech Detection in Vietnamese With a Unified Text-to-Text Transformer Model <https://aclanthology.org/2024.findings-acl.355.pdf>

Are Decoder-Only Language Models Better than Encoder-Only Language Models in Understanding Word Meaning? <https://aclanthology.org/2024.findings-acl.967.pdf>

ViGPTQA - State-of-the-Art LLMs for Vietnamese Question Answering: System Overview, Core Models Training, and Evaluations <https://aclanthology.org/2023.emnlp-industry.70.pdf>

ViGLUE: A Vietnamese General Language Understanding Benchmark and Analysis of Vietnamese Language Models <https://aclanthology.org/2024.findings-naacl.261.pdf>